Started on	Sunday, 24 July 2022, 4:35 PM
State	Finished
Completed on	Sunday, 24 July 2022, 4:35 PM
Time taken	15 secs
Grade	**0.00** out of 117.00 (**0**%)

Question **1**

Not answered

Marked out of 1.00

Flag question

Which of the following is associated with antidepressant induced hyponatraemia?

Select one:

- Age group 30-40
- Cold weather
- Normal renal function
- Low body weight
- Male sex

Your answer is incorrect.

Hyponatremia generally develops during the first month of therapy (3 to 120 days) with an incidence of 4.7/1000 treated patients per year. It is reversible between 2 and 28 days after the suspension of the SSRI. Risk factors include old age, low body mass index, concomitant use of diuretics, and female sex. (Br J Clin Pharmacol. 1999; 47:211-7.)

The correct answer is: Low body weight

Question **2**

Not answered

Marked out of 1.00

Flag question

A patient who is on an established antidepressant treatment with a SSRI takes over the counter cough preparations. Which of the following is likely to result due to an interaction between the medications?

Select one:

- neuroleptic malignant syndrome
- anticholinergic syndrome
- hypertensive syndrome
- cardiotoxicity
- serotonin syndrome

Your answer is incorrect.

The symptoms of the central serotonin syndrome include mental status changes, agitation, myoclonus, hyperreflexia, tremor, incoordination, headache, diaphoresis, hyperthermia, nausea, diarrhea, tachycardia, autonomic instability, cardiovascular collapse, and rarely, death. It is thought to be caused

by excessive secretion of serotonin from 5-HT1A receptors in the brain stem and spinal cord. It is commonly precipitated by drug-drug interactions between MAOIs and SSRIs/TCAs, but can occur between SSRIs and heterocyclic antidepressants and meperidine, triptans or dextromethorphan (in over-the-counter cough medications). Specifically, the combination of any SSRI, SNRI, or clomipramine (serotonergic medications) with any MAOI can cause this syndrome, and is the reason for a mandatory 1 week washout when changing from an SSRI (5 weeks for fluoxetine) to an MAOI.

The correct answer is: serotonin syndrome

Question **3**

Not answered

Marked out of 1.00

Flag question

A 32 year old man has been admitted to a medium secure forensic unit. 6 weeks after admission, he is observed to have tremors, ataxia, twitching, restlessness, diarrhoea, vomiting and polyuria. Within few days he is observed to have fluctuating alertness. The most likely diagnosis is

Select one:

- serotonin syndrome
- water intoxication
- anticholinergic syndrome
- acute dystonia
- tyramine reaction

Your answer is incorrect.

Major neurological symptoms of serotonin syndrome include myoclonus, tremors, chills, rigidity and hyperreflexia.

The correct answer is: serotonin syndrome

Question **4**

Not answered

Marked out of 1.00

Flag question

Which of the following has some evidence to support the notion of being the least cardiotoxic of all SSRIs?

Select one:

- Citalopram
- Sertraline
- Paroxetine
- Venlafaxine
- Fluoxetine

Your answer is incorrect.

Sertraline does not act as a stimulant or a sedative, has no anticholinergic activity and is not cardiotoxic; as a result, a large trial using sertraline in post-MI depression was undertaken, which re-affirmed the safety profile of sertraline, though clinical efficacy was not very impressive in this study.

The correct answer is: Sertraline

Question **5**

Not answered

Marked out of 1.00

Flag question

Mr. Z suffers from erectile dysfunction due to antidepressants and he was started on another medication to counteract this side effect. He develops Blue tinged vision. The most likely offending agent is;

Select one:

- Yohimbine
- Selegiline
- Bupropion
- Sildenafil citrate
- Agomelatine

Your answer is incorrect.

Sildenafil is effective at improving erectile dysfunction in men and in improving sexual function in women taking antidepressants, especially SSRIs. Some sildenafil users have complained of seeing everything 'tinted blue' also called as cyanopsia. The most common side effects of sildenafil use included headache, flushing, dyspepsia, nasal congestion, impaired vision including photophobia.

The correct answer is: Sildenafil citrate

Question **6**

Not answered

Marked out of 1.00

Flag question

Which of the following drugs can increase the teratogenic effects of carbamazepine?

Select one:

- Fluoxetine
- Lamotrigine
- Topiramate
- Buspirone
- Sodium valproate

Your answer is incorrect.

Both valproate and carbamazepine can synergistically increase the risk of neural tube defects.

The correct answer is: Sodium valproate

Question **7**

Not answered

Marked out of 1.00

Flag question

Which of the following is an example of idiosyncratic reaction?

Select one:

- Renal damage due to lithium
- Tardive dyskinesia due to antipsychotics
- SSRI discontinuation syndrome
- Steven Johnson syndrome on treatment with lamotrigine
- Lithium toxicity following a sepsis

Your answer is incorrect.

Steven Johnson syndrome is an idiosyncratic reaction; that could be life threatening. It is a drug-induced cutaneous reaction. It is associated with short-term therapy with phenytoin, phenobarbital, lamotrigine and carbamazepine. The period of increased risk is largely confined to the first eight weeks of treatment.

The correct answer is: Steven Johnson syndrome on treatment with lamotrigine

Question **8**

Not answered

Marked out of 1.00

Flag question

Which one among the following is a common cardiovascular side effect of clozapine?

Select one:

- Cardiomyopathy
- Tachycardia
- Thromboembolism
- Pericardial effusion
- Myocarditis

Your answer is incorrect.

Tachycardia is very common in early stages of treatment but usually benign. Tachycardia if persistent at rest and associated with fever, hypotension or chest pain may indicate myocarditis.

The correct answer is: Tachycardia

Question **9**

Not answered

Marked out of 1.00

Flag question

Which of the following is a side effect of lithium often noted even when the therapeutic levels in the serum are within normal limits?

Select one:

- ECG changes like those seen in hyperkalemia
- Peripheral limb oedema
- Thirst and polyuria
- EEG changes indicative of spike wave activity
- Left shift in White Blood Cell count

Your answer is incorrect.

Prior to starting lithium treatment, patients should be evaluated for renal, thyroid, and cardiac functions, as there are reports of acute changes in cardiac functioning. Lithium treatment can result in sick sinus syndrome. A baseline electrocardiograph (ECG) is beneficial to monitor any changes after the initiation of therapy (Managing Adverse Effects of Mood Stabilizers. Retrieved from http://primarypsychiatry.com/managing-adverse-effects-of-mood-stabilizers/). Lithium may unmask or aggravate Brugada syndrome, a hereditary disease of the cardiac sodium channel with characteristic ECG changes (right bundle branch block and ST segment elevation in right precordial leads), which may lead to cardiac arrest or sudden death. Lithium should not be administered to patients with Brugada syndrome or a family history of Brugada syndrome. Otherwise, cardiac adverse events are

rare with lithium treatment and result in T wave inversion or flattening (similar to hypokalemia). Thirst /polyuria are common side-effects, occurring in 2/3rd of patients in some case-series, despite relatively low levels of lithium. Limb oedema is a very rare side effect. (http://www.primarypsychiatry.com/aspx/articledetail.aspx?articleid=1138)

The correct answer is: Thirst and polyuria

Question **10**

Not answered

Marked out of 1.00

Flag question

Which of the following is NOT a known side effect of Sodium Valproate?

Select one:

- Red cell hypoplasia
- Leucopenia
- Weight gain
- Tremor
- Thrombocytosis

Your answer is incorrect.

Sodium valproate can cause leucopenia, thrombocytopenia and red cell hypoplasia

The correct answer is: Thrombocytosis

Question **11**

Not answered

Marked out of 1.00

Flag question

Which of the following psychotropic drugs is affected by acetylator (slow/fast) status?

Select one:

- Oxazepam
- Galantamine
- Chlorpromazine
- Lithium
- Nitrazepam

Your answer is incorrect.

- Acetylation is a phase 2 nonmicrosomal conjugation reaction.
- Drugs such as isoniazid, hydralazine, procainamide, some sulfonamides, sulfasalazine, nitrazepam, and caffeine undergo acetylation.
- The enzyme group responsible are the N-acetyltransferases.
- There are two different isozymes in humans, encoded by the genes NAT1 and NAT2.
- Based on the polymorphisms of NAT1 and NAT2, humans can be fast, intermediate, or slow acetylators.
- 60% of Caucasians and 10% to 20% of Asians are slow acetylators.

- Slow acetylation contributes to prolonged half-life of the drug. This increases the risk of side effects such as peripheral neuropathy (isoniazid), lupus syndrome (hydralazine and procainamaide), allergic reactions and hemolysis (sulfonamides), drowsiness, hangover (nitrazepam), and gastrointestinal effects (sulfasalazine).
- Ref: Kant A, Bilmen J, Hopkins PM. Adverse drug reactions. In Pharmacology and Physiology for Anesthesia 2019 Jan 1 (pp. 130-143). Elsevier.

The correct answer is: Nitrazepam

Question **12**

Not answered

Marked out of 1.00

Flag question

Which drug is most likely to cause weight loss and anorexia?

Select one:

- Carbamazepine
- Topiramate
- Sodium valproate
- Gabapentin
- Lamotrigine

Your answer is incorrect.

Topiramate is an anticonvulsant drug that is associated with severe/persistent loss of appetite and significant weight loss.

The correct answer is: Topiramate

Question **13**

Not answered

Marked out of 1.00

Flag question

Which of the following drugs is associated with nephrolithiasis?

Select one:

- Topiramate
- Lithium
- Valproate
- Carbamazepine
- Quetiapine

Your answer is incorrect.

Topiramate is associated with renal stone formation.

The correct answer is: Topiramate

Question **14**

Which of the following antidepressants has a potential to cause dependence?

Not answered

Marked out of 1.00

Flag question

Select one:

- Fluvoxamine
- Tranylcypromine
- Escitalopram
- Moclobemide
- Phenelzine

Your answer is incorrect.

Tranylcypromine has an amphetamine-like structure that renders it certain degree of addictive potential.

The correct answer is: Tranylcypromine

Question **15**

Not answered

Marked out of 1.00

Flag question

Which of the following antidepressants can prolong the QTc interval in normal doses?

Select one:

- Mirtazapine
- Duloxetine
- Fluoxetine
- Trazodone
- Reboxetine

Your answer is incorrect.

Maudsley Prescribing Guidelines. 9th ed. 224-225

The correct answer is: Trazodone

Question **16**

Not answered

Marked out of 1.00

Flag question

Commonest reported side effect of lithium includes

Select one:

- Pedal oedema
- EEG changes
- Thyroid toxicity
- Tremors
- Weight loss

Your answer is incorrect.

Gastrointestinal side effects (nausea, altered taste) and tremors are the most common side effects seen with lithium therapy.

The correct answer is: Tremors

Question **17**

Not answered

Marked out of 1.00

Flag question

A 65-year-old man developed cardiac arrhythmias on dosulepin. What is the next course of action?

Select one:

- Reduce dose
- Refer to a cardiologist
- Wait and watch
- Withdraw the drug immediately
- Repeat ECG after one week

Your answer is incorrect.

Dosulepin is a type of tricyclic antidepressant (TCA). Tricyclics have established arrhythmogenic activity, which arises as a result of a potent blockade of cardiac sodium channels and variable activity at potassium channels. Therefore, they should be avoided completely in patients with a risk of serious arrhythmias and should be withdrawn if they are already on it. Lofepramine for some reasons seems to lack the arrhythmogenicity of other TCAs.

The correct answer is: Withdraw the drug immediately

Question **18**

Not answered

Marked out of 1.00

Flag question

The risk factor for clozapine induced neutropenia is

Select one:

- White Caucasians
- Increased white cell disorder
- Young age
- Polypharmacy
- High dose

Your answer is incorrect.

The greatest reported risk of developing neutropenia or agranulocytosis with clozapine is during the first six months of treatment. Risk factors include being Afro-Caribbean (77% increase in risk) and young (17% decrease in risk per decade increase in age), and having a low baseline white cell count (WCC). This risk is not dose-related.

The correct answer is: Young age

Question **19**

Not answered

Marked out of 1.00

A drug that is less likely to be associated with metabolic syndrome than the others in the list is

Select one:

- chlorpromazine

Flag question

- olanzapine
- clozapine
- ziprasidone
- risperidone

Your answer is incorrect.

Therapy with clozapine and olanzapine is associated with an average increase in body weight of 9.9 pounds and 9.2 pounds, respectively, while those treated with risperidone or ziprasidone gained an average of 4.7 and 0.9 pounds, respectively. Ziprasidone has less propensity to cause metabolic syndrome.

The correct answer is: ziprasidone

Question **20**

Not answered

Marked out of 1.00

Flag question

Which of the following antidementia drugs undergo breakdown by cholinesterase itself with no hepatic involvement in metabolism?

Select one:

- Rivastigmine
- Donepezil
- Galantamine
- Memantine
- Tacrine

Your answer is incorrect.

Rivastigmine undergoes no hepatic conversion.

The correct answer is: Rivastigmine

Question **21**

Not answered

Marked out of 1.00

Flag question

Phenelzine has been shown to be superior to placebo in the treatment of

Select one:

- Arachnophobia
- Zoophobia
- Agoraphobia
- Social phobia
- Obsessional symptoms

Your answer is incorrect.

The monoamine oxidase inhibitors (MAOIs) have shown superior efficacy to other antidepressants in the the treatment of generalized social phobia. Phenelzine in particular, has extensive placebo-controlled trial evidence in support of its use in social anxiety with nearly 67% patients showing

significant improvement during acute treatment.

The correct answer is: Social phobia

Question **22**

Not answered

Marked out of 1.00

Flag question

Which one among the following statements about atypical antipsychotic drugs is incorrect?

Select one:

- Atypical antipsychotic drugs have relatively high serotonin HT-2A to dopamine D-2 receptor binding ratios
- The extrapyramidal safety of atypical antipsychotic drugs depends on the affinity for nigrostriatal dopamine D-2 receptors
- Relatively high 5HT-2A to dopamine D-2 binding ratios of atypical antipsychotic drugs may be responsible for the efficacy and safety advantages.
- Atypical antipsychotic drugs have less affinity for nigrostriatal than typical antipsychotic drugs
- The efficacy of antipsychotic drugs appear to be determined by the affinity for nigrostriatal dopamine receptors

Your answer is incorrect.

The efficacy of antipsychotic drugs appears to be determined by the affinity for mesolimbic dopamine receptors and not by the affinity for nigrostriatal receptors.

The correct answer is: The efficacy of antipsychotic drugs appear to be determined by the affinity for nigrostriatal dopamine receptors

Question **23**

Not answered

Marked out of 1.00

Flag question

Which of the following mood stabilisers is not recommended for prophylaxis of bipolar affective disorder?

Select one:

- Lithium
- Carbamazepine
- Topiramate
- Valproate
- Lamotrigine

Your answer is incorrect.

There are no large trials indicating the usefulness of topiramate in BPAD. Current evidence is sparse and limited.

The correct answer is: Topiramate

Question **24**

Not answered

Which of the following medications used to treat depression can directly influence gene transcription via nuclear receptors?

Marked out of 1.00

Flag question

Select one:

- L tryptophan
- Phenelzine
- Fluoxetine
- Imipramine
- Tri-iodothyronine

Your answer is incorrect.

As triiodothyronine acts via nuclear receptors, it can influence gene transcription directly.

The correct answer is: Tri-iodothyronine

Question **25**

Not answered

Marked out of 1.00

Flag question

A 32-year-old woman with schizophrenia is taking an antipsychotic medication. She has amenorrhoea and galactorrhoea. This is due to blockade of D2 receptors in which of the following pathways?

Select one:

- Mesocortical pathway
- Tuberoinfundibular pathway
- Mesolimbic pathway
- Papez circuit
- Niagrostriatal pathway

Your answer is incorrect.

This woman is experiencing features of hyperprolactinaemia that includes amenorrhoea, galactorrhoea and loss of sexual interest. It occurs due to D2 blockade in the tuberoinfundibular pathway.

The correct answer is: Tuberoinfundibular pathway

Question **26**

Not answered

Marked out of 1.00

Flag question

Which of the following drugs can alleviate the symptoms of drug craving by means of a partial agonistic action?

Select one:

- Methadone
- Acamprosate
- Disulfiram
- Bupropion
- Varenicline

Your answer is incorrect.

Varenicline is a partial nicotinic agonist. It alleviates the symptoms of nicotine craving and withdrawal

through its agonist activity while inhibiting the effects of repeated nicotine exposure by its antagonist activity. http://www.medscape.com/viewarticle/559955_3

The correct answer is: Varenicline

Question **27**

Not answered

Marked out of 1.00

Flag question

Gabapentin binds to which of the following receptors?

Select one:

- GABA-B receptor
- Voltage-dependent calcium channel
- Omega benzodiazepine receptor unit
- GABA-A receptor
- NMDA receptor

Your answer is incorrect.

To date, the mechanism for the analgesic and anticonvulsant effects of gabapentin are not known. Despite its name, Gabapentin does not bind to GABA receptors, does not affect GABA uptake and does not inhibit GABA transaminase. Gabapentin passes through blood-brain-barrier to bind to the a2d1 (alpha2delta1)-type voltage-gated calcium channels.

The correct answer is: Voltage-dependent calcium channel

Question **28**

Not answered

Marked out of 1.00

Flag question

Which of the following is a prodrug whose metabolite is used as an individual compound therapeutically?

Select one:

- Citalopram
- Zopiclone
- Fluoxetine
- Risperidone
- Haloperidol

Your answer is incorrect.

Paliperidone is 9-OH-risperidone.

Citalopram is not a prodrug - escitalopram is the therapeutically active S-enantiomer of citalopram.

The correct answer is: Risperidone

Question **29**

Not answered

Marked out of 1.00

Which antipsychotic has a longer therapeutic effect attributable to its active metabolite?

Select one:

- Amisulpride

Flag question

- Risperidone
- Chlorpromazine
- Haloperidol
- Sulpride

Your answer is incorrect.

The apparent half-life of risperidone is 3 hours while the apparent half-life of 9-hydroxyrisperidone (also known as paliperidone) is about 21 hours (in extensive metabolizers). The pharmacokinetics of risperidone and 9-hydroxyrisperidone combined, after single and multiple doses, were similar in extensive and poor metabolizers, with an overall mean elimination half-life of about 20 hours (RISPERDAL M-TAB - Food and Drug Administration, http://www.fda.gov/ohrms/dockets/dockets/06p0245/06P-0245-EC1-Attach-1.pdf).

The correct answer is: Risperidone

Question **30**

Not answered

Marked out of 1.00

Flag question

Which of the following measures can help reduce lithium-induced tremors?

Select one:

- Drug holidays
- Single large dose
- Taking lithium after food
- Smaller, more frequent doses
- Change time of administration to mornings only

Your answer is incorrect.

Administer the lithium in smaller, more frequent doses. Propranolol (30 to 160 mg/d) may also be helpful.

The correct answer is: Smaller, more frequent doses

Question **31**

Not answered

Marked out of 1.00

Flag question

A psychiatric trainee has started lamotrigine in addition to a second mood stabiliser for a lady with a relapse of bipolar affective disorder. The pharmacist is concerned that the second mood stabiliser will increase possibility of toxicity of the lamotrigine. The second mood stabiliser is most likely to be

Select one:

- Olanzapine
- Sodium valproate
- Lithium
- Risperidone
- Carbamazepine

Your answer is incorrect.

Sodium valproate can inhibit the breakdown of lamotrigine and so can result in higher plasma levels of lamotrigine. Valproate also increases the risk of allergic rash seen with lamotrigine.

The correct answer is: Sodium valproate

Question **32**

Not answered

Marked out of 1.00

Flag question

Mrs. X suffers from depressive disorder and has been on paroxetine for the last 8 weeks. She has not made any improvement and stopped taking her medication. She is now complaining of dizziness, insomnia and shock like sensations. The most likely diagnosis is

Select one:

- SSRI related toxicity
- Serotonin syndrome
- Neuroleptic malignant syndrome
- Teratogenicity
- SSRI discontinuation syndrome

Your answer is incorrect.

Explanation:

SSRI discontinuation symptoms may include flu-like symptoms, 'shock-like' sensations, dizziness (most common), insomnia, excessive dreaming, irritability, weakness, lethargy/fatigue, GI symptoms, and tearfulness. The onset of symptoms is usually within days (or 3-5 drug half-lives) of stopping treatment.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 143.

The correct answer is: SSRI discontinuation syndrome

Question **33**

Not answered

Marked out of 1.00

Flag question

Julie is a 29-year-old woman who has recently started taking combined oral contraceptive pills. Which of the following medication could decrease the effectiveness of the contraception?

Select one:

- St. John's wort
- Fluoxetine
- Lithium
- Clozapine
- Amitriptyline

Your answer is incorrect.

Explanation:

Psychiatric drugs that are CYP3A4 inducers (e.g. carbamazepine, St John's Wort, phenytoin, topiramate) reduce the bioavailability and hence contraceptive efficacy of combined hormonal contraceptives, the progesterone-only pill, and the subdermal implant both during use and for up to 28

days after stopping enzyme-inducer treatment. Thus, patients on enzyme inducers should be encouraged to use intrauterine contraception or injectables.

Ref: Taylor DM, Gaughran F, Pillinger T. The Maudsley Practice Guidelines for Physical Health Conditions in Psychiatry. 2021. p. 314.

Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 855.

The correct answer is: St. John's wort

Question **34**

Not answered

Marked out of 1.00

Flag question

A 69-year-old man on warfarin has come back from a holiday in France when he developed some psychiatric symptoms. He was also taking one of the following drugs. Which one of the above drugs would lead you to send him for an urgent INR?

Select one:

- Clozapine
- St. John's wort
- Olanzapine
- Lithium
- Venlafaxine

Your answer is incorrect.

Explanation:

St. John’s wort (SJW) is a potent inducer of intestinal and hepatic CYP3A4, CYP2C9, CYP2C19, CYP2E1, and intestinal p-glycoprotein. Hyperforin is responsible for this effect. The hyperforin content of SJW preparations varies 50-fold, which will result in a different propensity for drug interactions between brands. Studies have shown that SJW significantly reduces plasma concentrations of warfarin, hormonal contraceptives, digoxin, and indinavir (used in the treatment of HIV).

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 418.

The correct answer is: St. John's wort

Question **35**

Not answered

Marked out of 1.00

Flag question

Which of the following is correct with respect to first-order elimination kinetics?

Select one:

- Constant amount of drug is eliminated per unit of time
- Very few drugs undergo first order metabolism
- Steady state concentration of the drug is doubled by doubling the administration rate
- Clearance depends on concentration of the drug in plasma
- Most commonly seen when metabolic enzymes are saturated

Your answer is incorrect.

Explanation:

Constant fraction, not amount is eliminated per unit time if a drug undergoes first order elimination. Hence, clearance is independent of drug concentration in plasma. Doubling the administration rate will double the steady-state plasma concentration.

Ref: Laracuente ML, Marina HY, McHugh KJ. Zero-order drug delivery: State of the art and future prospects. Journal of Controlled Release. 2020 Nov 10;327:834-56.

The correct answer is: Steady state concentration of the drug is doubled by doubling the administration rate

Question **36**

Not answered

Marked out of 1.00

Flag question

Which of the following statements about protein binding is incorrect?

Select one:

- Tissue binding of a drug cannot be directly measured
- The distribution of a drug is not affected by the degree of protein binding
- Plasma protein binding is usually reversible
- Pregnancy can affect protein binding of drugs
- Displacement bound fraction of a drug from plasma proteins may be a basis of drug interactions.

Your answer is incorrect.

The distribution of a drug depends on how protein bound it is. When in the blood, many drugs are bound to circulating plasma proteins, predominantly albumin but also globulins and glycoproteins. It is the unbound fraction of the drug (free fraction) that can be active i.e., bind to receptors, pass across blood brain barrier, etc. Therefore, changes in protein binding can have profound effects on the availability of the drugs. Most psychotropic drugs are basic, and they may bind to, for example, alpha-1 acid glycoprotein and lipoproteins. Plasma protein binding is usually reversible (not covalent).

The correct answer is: The distribution of a drug is not affected by the degree of protein binding

Question **37**

Not answered

Marked out of 1.00

Flag question

All of the following drugs reduce renal clearance of lithium except

Select one:

- Theophylline
- Thiazides
- Ibuprofen
- Diclofenac
- Frusemide

Your answer is incorrect.

Xanthine diuretics such as theophylline increase the renal clearance of lithium.

The correct answer is: Theophylline

Question **38**

Not answered

Marked out of 1.00

Flag question

Regarding the pharmacokinetic properties of MAOIs, which one of the following statements is true?

Select one:

- They are mainly excreted unchanged by the kidney
- They have a short elimination half life
- Irreversible non selective MAOIs are highly water soluble
- The acetylation status is polygenetically inherited.
- About 90% of white people are fast acetylators

Your answer is incorrect.

Irreversible non-selective MAOIs are highly lipid soluble, rapidly absorbed and widely distributed. They have a short half-life (1 hour for phenelzine and 2 hours for Tranylcypromine). They are extensively metabolised by the liver. Only 1-2% is excreted unchanged in the urine. About 50% of white people are a higher proportion of Orientals are fast acetylators. They are ontogenetically inherited acetylator status.

The correct answer is: They have a short elimination half life

Question **39**

Not answered

Marked out of 1.00

Flag question

Amisulpride is especially suitable in which of the following situations?

Select one:

- To treat a pregnant woman with psychosis
- To treat a patient with psychosis and electrolyte imbalance
- To treat adolescent onset schizophrenia
- To treat an elderly woman with psychosis and osteoporosis
- To treat a patient with psychosis and hepatic dysfunction

Your answer is incorrect.

Amisulpride undergoes little, if any, hepatic metabolism, making it a favourite drug to treat psychosis in patients with hepatic dysfunction

The correct answer is: To treat a patient with psychosis and hepatic dysfunction

Question **40**

Not answered

Marked out of 1.00

Flag question

When administered orally, the benzodiazepine with the shortest elimination half-life is

Select one:

- Nitrazepam
- Alprazolam
- Triazolam
- Chlordiazepoxide
- Lorazepam

Your answer is incorrect.

Triazolam has a half-life for 2 hours and has the shortest half-life among those listed.

The correct answer is: Triazolam

Question **41**

Not answered

Marked out of 1.00

Flag question

The rate limiting step in dopamine metabolism is catalysed by

Select one:

- Tyrosine hydroxylase
- Monoamine oxidase
- Tryptophan hydroxylase
- Catechol O Methyl transferase
- Dopamine hydroxylase

Your answer is incorrect.

The rate-limiting step in dopamine metabolism is catalysed by the enzyme Tyrosine hydroxylase. Tyrosine is converted into DOPA, using Tyrosine hydroxylase. DOPA is then converted into dopamine.

The correct answer is: Tyrosine hydroxylase

Question **42**

Not answered

Marked out of 1.00

Flag question

Having more than 2 copies of fully active CYP450 enzyme alleles will result in which of the following phenotype?

Select one:

- Ultrarapid metaboliser
- Absent metabolism
- Extensive metaboliser
- Intermediate metaboliser
- Poor metaboliser

Your answer is incorrect.

Having normal 2 copies will produce extensive metabolism. Having 1 copy will result in intermediate metabolism, while poor metabolizers do not have any active enzyme allele at all. Having more than 2 copies results in being ultrarapid metabolizer that may lead to failure of drug effect at normal dosage.

Ref: https://www.ncbi.nlm.nih.gov/books/NBK100662/

The correct answer is: Ultrarapid metaboliser

Question **43**

Which drug increases the levels of lamotrigine?

Not answered

Marked out of 1.00

Flag question

Select one:

- Carbamazepine
- Modafanil
- Topiramate
- Promethazine
- Valproate

Your answer is incorrect.

Concomitant administration of sodium valproate reduces lamotrigine total clearance by approximately 21% and increases elimination half-life. The most probable explanation for this effect is hepatic competition between VPA and LTG for glucuronidation. Adding lamotrigine to valproate decreases the steady-state levels of valproate while the steady-state levels of lamotrigine increase approximately twofold. In this case, the starting dose of lamotrigine should be lowered, and the titration made slowly.

The correct answer is: Valproate

Question **44**

Not answered

Marked out of 1.00

Flag question

Which of the following is a safe drug to use in a woman who is taking oral contraceptive pills?

Select one:

- Carbamazepine
- Topiramate
- Phenytoin
- Valproate
- phenobarbitone

Your answer is incorrect.

Patients taking oral contraceptive steroids (OCS) are known to suffer contraceptive failure while taking anticonvulsants such as phenobarbitone, phenytoin and carbamazepine though valproate has been traditionally used as a safer alternative. Note that even Topiramate in doses higher than 200mg may make combined oral contraceptive pills less effective by reducing the blood level of oestrogen. There are no interactions between the combined oral contraceptive pill and sodium valproate, vigabatrin, lamotrigine, gabapentin, tiagabine, levetiracetam and the benzodiazepines (Crawford, 2000).

The correct answer is: Valproate

Question **45**

Not answered

Marked out of 1.00

Flag question

Mr. Y is a 45-year-old gentleman treated on tranylcypromine for major depression without much benefit. He would like his medication to be changed to paroxetine. The best course of action would be

Select one:

- Withdraw and wait for 4 weeks before starting new drug
- Withdraw and wait for 3 weeks before starting new drug
- Withdraw and wait for 2 weeks before starting new drug

- Withdraw and wait for 1 week before starting new drug
- Withdraw and wait for 5 weeks before starting new drug

Your answer is incorrect.

When you switch from MAOIs to SSRIs, withdraw and wait for 2 weeks before starting a new drug. This will help to prevent the incidence of serotonin syndrome.

The correct answer is: Withdraw and wait for 2 weeks before starting new drug

Question **46**

Not answered

Marked out of 1.00

Flag question

A constant amount, not a fraction, of the drug compound is cleared per unit time. This is called;

Select one:

- Phase 1 metabolism
- Phase 2 metabolism
- Zero order kinetics
- Bioequivalence
- First order kinetics

Your answer is incorrect.

When a constant fraction of drug is cleared per unit time, it is called as first order kinetics. Thus, if the ingested drug increases in quantity, more of it will be eliminated. Most psychotropic drugs follow first order kinetics. In zero order kinetics, a constant amount, not the fraction, of a drug is cleared per unit time. This means that irrespective of the amount of drug in plasma or dose of drug administered, the body clears only a fixed unit of the drug. Furhter increases in dose might result in serious toxicity in this case.

The correct answer is: Zero order kinetics

Question **47**

Not answered

Marked out of 1.00

Flag question

Val has been started on a hypnotic medication following an admission for a severe depressive episode. Since starting the medication, she complains of a metallic taste on her tongue. The offending medication is most likely to be

Select one:

- Oxazepam
- Zopiclone
- Nitrazepam
- Zolpidem
- Temazepam

Your answer is incorrect.

Zopiclone has a unique side effect of producing unpleasant, metallic taste.

The correct answer is: Zopiclone

Question **48**

Not answered

Marked out of 1.00

Flag question

The enzymes (pair of) involved in the metabolism of serotonin is

Select one:

- MAO-A
- GABA Decarboxylase
- MAO-B
- Catechol-o-methyl transferase
- Choline acetyl transferase

Your answer is incorrect.

The breakdown enzyme involved in the metabolism of serotonin is MAO (Preferentially MAO-A)

The correct answer is: MAO-A

Question **49**

Not answered

Marked out of 1.00

Flag question

Mr. Y was diagnosed with HIV almost 2 years ago. He is now presenting with cognitive impairment. Which of the following cells is least likely to be affected?

Select one:

- Lymphocytes
- None of the above
- Neurones
- Macrophages
- Microglia

Your answer is incorrect.

Theoretically all the main cell types of the CNS, astrocytes, oligodendrocytes, neurons, perivascular macrophage and microglia, can be infected by HIV-1 since they possess the receptors and/or co-receptors for HIV-1 entry, but only the latter two are the most commonly infected cells by HIV-1.

Most studies have indicated an absence of in vivo infection in neurons. It is unclear whether detection of infected neurons is complicated by the loss of the infected neuronal populations.

(Excerpt from Ghafouri et al., Retrovirology 2006, 3:28).

The correct answer is: Neurones

Question **50**

Not answered

Marked out of 1.00

Regional cerebral blood flow is affected in schizophrenia. Which of the following area is most likely to be affected?

Select one:

- Left medial temporal

Flag question

- Prefrontal cortex
- Raphe nuclei
- Basal ganglia
- Bilateral parietal

Your answer is incorrect.

Hypofrontality has been repeatedly demonstrated.

In schizophrenia:
Hypoperfusion: frontal lobes and frontal cortex, anterior and medial cingulate gyri, and parietal lobes.
Hyperperfusion: cerebellum, brainstem and thalamus.

The correct answer is: Prefrontal cortex

Question **51**

Not answered

Marked out of 1.00

Flag question

The post-mortem examination done on an elderly gentleman who suffered from a slowly progressive non-amnesic cognitive impairment, disinhibition and pill-rolling tremors reveals ubiquitin inclusions. Which one among the following genes is most likely to reveal a mutation?

Select one:

- Progranulin
- Prion protein
- Neurexin
- Presenilin-2
- Dysbindin

Your answer is incorrect.

This is a case of frontotemporal dementia (FTD). FTD is caused by mutations in progranulin (PGRN) especially in familial FTD. PGRN is located on chromosome 17q21.31.

The correct answer is: Progranulin

Question **52**

Not answered

Marked out of 1.00

Flag question

The density of D2 receptors seen in children before 5 years of age is greater than adult levels. Their regression to adult levels is reached during the second decade. The developmental process through which this is achieved is called

Select one:

- Downregulation
- Pruning
- Desensitisation
- Waning
- Habituation

Your answer is incorrect.

Synaptic 'pruning' refers to the natural elimination of unnecessary synapses that normally occurs in adolescence, making the brain more organised and efficient.

The correct answer is: Pruning

Question **53**

Not answered

Marked out of 1.00

Flag question

A disturbance in cerebral torque is proposed as the underlying pathological change in which of the following disorders?

Select one:

- OCD
- Autism
- Grand mal epilepsy
- PTSD
- Psychosis

Your answer is incorrect.

'Cerebral torque' refers to opposing right-left asymmetries of frontal and parieto-occipital regions of the brain.

Crow's hypothesis and observations suggest that a primary disturbance in psychosis is in brain structure, and this is related to cerebral asymmetry in the form of torque.

The correct answer is: Psychosis

Question **54**

Not answered

Marked out of 1.00

Flag question

Which of the following is the most common opportunistic infection of CNS in patients with AIDS?

Select one:

- Cryptococcal meningitis
- Neurocysticercosis
- Toxoplasma gondii
- Cytomegalovirus infection
- Tuberculous meningitis

Your answer is incorrect.

Toxoplasmic encephalitis (TE) is caused by the protozoan Toxoplasma gondii. The disease occurs almost exclusively because of reactivation of latent tissue cysts. Primary infection occasionally is associated with acute cerebral or disseminated disease. Seroprevalence varies substantially among different communities (e.g., approximately 15% in the United States and 50%--75% in certain European countries). Prior to the introduction of antiretroviral drugs, the 12-month incidence of TE was approximately 33%. The incidence has decreased substantially with the initiation of ART. Clinical

disease is rare among patients with CD4+ T lymphocyte counts more than 200 cells/μL. Primary infection occurs after eating undercooked meat containing tissue cysts or ingestion of oocysts that have been shed in cat feces and have sporulated in the environment (which requires at least 24 hours). No transmission of the organism occurs by person-to-person contact (Retrieved from http://www.cdc.gov/mmwr/preview/mmwrhtml/rr58e324a1.htm).

The correct answer is: Toxoplasma gondii

Question **55**

Not answered

Marked out of 1.00

Flag question

A mutation involving which of the following proteins is not associated with neurodegeneration?

Select one:

- Tubulin
- Superoxide dismutase
- Tau
- Synuclein
- Huntingtin

Your answer is incorrect.

Huntingtin is associated with Huntington disease; Tau proteins with FTD; Synuclein with Lewy body dementia and Parkinson's; SOD mutation is associated with MND (Amyotrophic Lateral Sclerosis). Tubulin mutations are accoiated with cancer syndromes mostly - recently an association has been demonstrated with pachygyria and lissencephaly - cortical folding disorders with no neurodegenerative features.

The correct answer is: Tubulin

Question **56**

Not answered

Marked out of 1.00

Flag question

Which of the following disorders is characterized by the presence of Hirano bodies and balloon cells (intracellular vacuoles)?

Select one:

- Fronto-temporal dementia
- Schizophrenia
- Sleeping sickness
- Shy-Drager syndrome
- Alzheimer's dementia

Your answer is incorrect.

Hirano bodies are bright eosinophilic intracytoplasmic inclusions which occur in a wide variety of conditions, especially amyotrophic lateral sclerosis and Parkinsonism-dementia complex, Alzheimer's disease, Pick's disease and 'normal' elderly individuals. In Pick's disease (a form of fronto-temporal dementia), there are two distinctive histologic features: swelling of neurons (resulting in "ballooned Pick cells") and an inclusion within the perikaryon, most often in layer II (resulting in a Pick body). Pick bodies are usually found in limbic, paralimbic, and ventral temporal lobe cortex. Both Hirano bodies and balloon cells are seen in fronto-temporal dementia

The correct answer is: Fronto-temporal dementia

Question **57**

Not answered

Marked out of 1.00

Flag question

Which of the following is a glucocorticoid receptor antagonist?

Select one:

- D-Cycloserine
- Clozapine
- Mifepristone
- Agomelatine
- Gabapentin

Your answer is incorrect.

Mifepristone is a glucocorticoid receptor antagonist. It is a blocker of type 2 glucocorticoid and progestin receptors, and the preliminary research findings suggest that it could effectively ameliorate symptoms of major depression.

The correct answer is: Mifepristone

Question **58**

Not answered

Marked out of 1.00

Flag question

The EEG findings of a patient shows multiple spike and wave activity. Identify the disease.

Select one:

- Absence seizure
- Creutzfeldt Jacob disease
- Subacute sclerosing panencephalitis
- Herpes encephalitis
- Myoclonic epilepsy

Your answer is incorrect.

Common EEG findings include rapid spike wave and poly spike wave.

The correct answer is: Myoclonic epilepsy

Question **59**

Not answered

Marked out of 1.00

Flag question

The substance that is secreted by hypothalamus and enhances food intake is

Select one:

- Neuropeptide Y
- Beta endorphins
- Neurotensin
- Vasopressin
- Cholecystokinin

Your answer is incorrect.

Food intake is regulated by various factors such as neuropeptide Y. Neuropeptide Y potently induces an increase in food intake, and simultaneously stimulates arginine-vasopressin (AVP) secretion in the brain (Retrieved from http://www.ncbi.nlm.nih.gov/pubmed/19766110).

The correct answer is: Neuropeptide Y

Question **60**

Not answered

Marked out of 1.00

Flag question

There is increased concentration in axon terminal prior to depolarisation with which of the following ions?

Select one:

- Calcium
- Nitrogen
- Sodium
- Hydrogen
- Potassium

Your answer is incorrect.

There is a negative resting membrane potential of around -70mV. It is maintained by the sodium pump, which actively transports sodium ions out of the cell and potassium ions into the cell. There is increased concentration of potassium ions in axon terminal prior to depolarization. An action potential is propagated by the depolarization spreading laterally to adjacent parts of the neurone. An action potential is initiated in the axon hillock when the synaptic signals received by the dendrites and soma are sufficient to raise the intracellular potential from -70 mV to the threshold potential of - 55mV. When this potential is reached, the Na+ channels present in the axon initial segment will open. This Na+ influx causes a rapid reversal of the membrane potential from -90 to +40 mV. When the membrane potential reaches +40mV, the Na+ channels close and the voltage-gated K+ channels open. K+ ions move out of the axon, and 'repolarizes' the membrane.

The correct answer is: Potassium

Question **61**

Not answered

Marked out of 1.00

Flag question

Which of the following is least likely to be correct with regard to REM latency and depression? (June 2009, Nov2009)

Select one:

- REM sleep latency reduction precedes onset of major depressive disorder
- Shortened REM latency is specific to depression
- Relatives of depressed patients have increased prevalence of short REM latency
- Reduced REM latency could persist even after successful treatment of depressive episode
- REM latency reduction predicts higher relapse risk

Your answer is incorrect.

Short REM latency is associated with increased risk of major depression beyond the familial risk associated with a depressed proband. Consistent with other heritable traits, both REM latency reduction and Slow Wave Sleep abnormalities have been observed preceding a first episode of depression in patients. But it has low predictive value as substantial number of those who have short REM latency, do not develop depression. Short REM latency is not specific to depression; it may be seen in PTSD, schizophrenia, eating disorders, etc

The correct answer is: Shortened REM latency is specific to depression

Question **62**

Not answered

Marked out of 1.00

Flag question

Occasional sleep spindles and K complexes occurs during which of the following stages of sleep?

Select one:

- Stage 4 n-REM sleep
- Stage 2 n-REM sleep
- Stage 1 n-REM sleep
- REM sleep
- Stage 3 n-REM sleep

Your answer is incorrect.

EEG during sleep is an important topic for the MRCPsych exams. Stage 0: Alpha activity (awake but relaxed) NREM Stage 1: Low amplitude, reduced alpha activity, low voltage theta activity NREM Stage 2: 2-7 Hz, Occasional sleep spindles and high voltage K complexes NREM Stage 3: Increased delta activity (20 to 50 percent)-large slow waves appear NREM Stage 4: Increased delta activity (more than 50 percent)-large slow waves predominate REM sleep: Asymmetrical saw tooth waves, occasional alpha rhythm

The correct answer is: Stage 2 n-REM sleep

Question **63**

Not answered

Marked out of 1.00

Flag question

The EEG waves with a frequency of 4.0-7.5 Hz are called

Select one:

- Mu
- Theta
- Beta
- Alpha
- Delta

Your answer is incorrect.

Theta rhythms vary in frequency between 4 and 8 Hz and are thought to be generated by many neurons firing synchronously in the hippocampus and the cortex.

The correct answer is: Theta

Question **64**

Not answered

Marked out of 1.00

Flag question

The wavelet band of EEG that is 8-12Hz in frequency is

Select one:

- Gamma
- Delta
- Beta
- Theta
- Alpha

Your answer is incorrect.

The EEG usually has the following frequency bands: Gamma (> 30Hz) BETA (13-30Hz), ALPHA (8-12 Hz), THETA (4-8 Hz), and DELTA (<4 HZ)

The correct answer is: Alpha

Question **65**

Not answered

Marked out of 1.00

Flag question

You are working in the memory disorders clinic. You would like to test various cognitive domains using MMSE. Which of the following is NOT tested in the MMSE?

Select one:

- Attention
- Comprehension
- Orientation
- Executive functions
- Concentration

Your answer is incorrect.

MMSE is based entirely on verbal assessment of memory and attention. It is insensitive to frontal executive dysfunction and visuospatial deficits.

The correct answer is: Executive functions

Question **66**

Not answered

Marked out of 1.00

Flag question

John has Huntington's disease and suffers from severe cognitive deficits. He recently had an EEG test. What findings can be expected in the report?

Select one:

- Focal delta activity
- 3Hz spike and wave activity
- 1-2 Hz sharp waves
- Flattened trace
- Spike and wave complexes

Your answer is incorrect.

EEG findings in Huntington's dementia includes a loss of alpha waves initially; but later the trace becomes flattened, especially in the frontal regions, as the dementia progresses. Ref: Scott, D.F., Heathfield, K.W.G., Toone, B. and Margerison, J.H., 1972. The EEG in Huntington's chorea: a clinical and neuropathological study. Journal of Neurology, Neurosurgery & Psychiatry, 35(1), pp.97-102.

The correct answer is: Flattened trace

Question **67**

Not answered

Marked out of 1.00

Flag question

Multiple deep white matter hyperintensities in neuroimaging studies are predominantly seen in older adults with

Select one:

- Obsessive compulsive disorder
- Bipolar disorder
- Autistic spectrum disorder
- Schizophrenia
- Depression

Your answer is incorrect.

"The association between major depressive disorder and increased prevalence of brain white-matter hyper intensities has been reported in elderly people (Krishnan et al, 1997; de Groot et al, 2000), but studies in younger patients have been inconclusive (Coffey et al, 1993; Lenze et al, 1999; Lyoo et al, 2002). In samples of elderly people with depression, such hyper intensities were associated with lower rates of response to antidepressant treatment (Hickie et al, 1997; Simpson et al, 1998; Steffens et al, 2002) as well as higher rates of relapse during follow-up (O'Brien et al, 1998; Yanai et al, 1998)". Excerpted from: Iosifescu DV, Renshaw PF, Lyoo IK, Lee HK, Perlis RH, Papakostas GI, Nierenberg AA, Fava M. Brain white-matter hyperintensities and treatment outcome in major depressive disorder. The British Journal of Psychiatry. 2006 Feb 1;188(2):180-5.

The correct answer is: Depression

Question **68**

Not answered

Marked out of 1.00

Flag question

The male: female ratio of pure frontal lobe degeneration is

Select one:

- 1 to 3
- 1 to 1
- 1 to 2
- 2 to 1
- 3 to 1

Your answer is incorrect.

Frontotemporal lobe degeneration is characterized by progressive neuronal loss involving the frontal and or temporal lobes. The signs and symptoms often manifest between the ages of 55 and 65. 15% of those with FTD also have MND i.e. Motor Neuron Disease (also called Amyotrophic Lateral Sclerosis,

ALS). MND with or without FTD is slightly more likely to be seen in men than women, but the difference is small. Population studies show nearly equal distribution by gender for pure frontal lobe degeneration and FTD without MND. Ref: http://www.ncbi.nlm.nih.gov/pmc/articles/PMC3932112/

The correct answer is: 1 to 1

Question **69**

Not answered

Marked out of 1.00

Flag question

ECG changes in Huntington's disease include an increase in the incidence of

Select one:

- Conduction abnormalities and bradycardia
- Tachycardia
- Atrial fibrillation
- Silent myocardial infarcts
- Ventricular ectopics

Your answer is incorrect.

There is increasing evidence that, in Huntington's, in addition to central nervous system pathology, peripheral tissues may also be affected. In particular, animal models of HD show significant cardiac dysregulation. Patients with HD show sympathovagal disturbance and increased rates of sudden cardiac death. Nearly 25% of HD patients show an abnormal ECG. Bradycardia, prolonged intraventricular conduction and unexplained QTc elevation were the most commonly observed changes. Ref: http://www.neurology.org/content/84/14_Supplement/P5.294

The correct answer is: Conduction abnormalities and bradycardia

Question **70**

Not answered

Marked out of 1.00

Flag question

A 42-year-old woman with no prior neurological or psychiatric history presents to you with discomfort in her legs, which prevents her from sleeping. The most probable diagnosis is

Select one:

- Serotonin syndrome
- Tardive dyskinesia
- Restless Leg Syndrome
- Neuroleptic malignant syndrome
- REM sleep disorder

Your answer is incorrect.

Restless leg syndrome: It is characterized by an unpleasant sensation inside the legs, occurring in the evenings, an urge to move the legs and relief by movement. It leads to repetitive movements of the legs, which disrupts sleep. The person may complain of daytime fatigue and difficulty in initiating sleep.

The correct answer is: Restless Leg Syndrome

Question **71**

Not answered

Marked out of 1.00

Flag question

A genetic disorder is inherited through recessive X-linked transmission. If the probability of the disease in males is 1/100, then the probability of the same disease presenting in females will be

Select one:

- 1/1000
- 1/ 10000
- 1/ 10000000
- 1/10
- 1 / 100

Your answer is incorrect.

According to Hardy Weinberg equilibrium, in populations where mating is random, allele frequencies are constant across generations, and can be mathematically quantified. Autosomal alleles: If the frequency of one allele is p, and the other is q, then the frequency of the homozygous genotypes will be p2 and q2 respectively and the frequency of the heterozygous genotype will be p*q. According to Hardy Weinberg rule, p2 +2pq +q2 = 1. X chromosome alleles: As females have 2 X chromosomes, and males have 1 X and 1 Y chromosome, the Hardy Weinberg expectations for an X-linked trait are the same as for an autosomal locus for females, but for males they are simply the allele frequencies. Let us assume that a locus has 2 alleles A and B, and the allele frequencies p and q. Then for females, the frequency of AA and BB are p2 and q2 respectively; and AB is given by 2pq. But for males, as only one copy of X is present, the frequency of A and B are p and q respectively. In the MCQ here, q is given as 1/100. In this case q2 is given by 1/10000.

The correct answer is: 1/ 10000

Question **72**

Not answered

Marked out of 1.00

Flag question

The mode of transmission of Fragile X syndrome is

Select one:

- x-linked dominant
- x-linked recessive
- autosomal recessive
- Disorder of unknown etiology
- autosomal dominant

Your answer is incorrect.

Fragile X syndrome is a common inherited cause of learning disability with an atypical X-linked dominant transmission. Penetrance is overall low, but greater in males than females.33% of heterozygous females can have some features of the phenotype; thus it is not a X-linked recessive illness; in addition, the inheritance pattern shows genetic anticipation phenomenon typical of trinucleotide repeat disorders. Ref: Garber KB, Visootsak J, Warren ST. Fragile X syndrome. European Journal of Human Genetics. 2008 Jun 1;16(6):666-72.

The correct answer is: x-linked dominant

Question **73**

Not answered

Marked out of 1.00

Flag question

Which one of the following is considered as a protective factor for Alzheimer's disease?

Select one:

- Previous head injury
- Family history of down's syndrome
- Parkinson's disease
- Apolipoprotein E2
- Women

Your answer is incorrect.

The presence of apolipoprotein e2 allele is a proven protective factor against Alzheimer's disease. Other putative protective factors include physical exercise, oestrogen replacement, higher premorbid intelligence and higher levels of education, though the size of the association for these factors are often small and inconsistent across studies. Ref: Reitz C, Brayne C, Mayeux R. Epidemiology of Alzheimer disease. Nature Reviews Neurology. 2011 Mar 1;7(3):137-52.

The correct answer is: Apolipoprotein E2

Question **74**

Not answered

Marked out of 1.00

Flag question

You are seeing a child with elfin facial features and hoarse voice. The most probable diagnosis is

Select one:

- Patau syndrome
- Williams syndrome
- Smith-Magenis syndrome
- Di George syndrome
- Down syndrome

Your answer is incorrect.

Clinical features of Williams's syndrome include hypercalcemia (50% of cases) with supravalvular aortic stenosis and unusual facies. Other features include growth retardation, 'elfin' facial features, hoarse voice, premature wrinkling and sagging of the skin, urinary tract abnormalities and mild to moderate Learning disability.

The correct answer is: Williams syndrome

Question **75**

Not answered

Marked out of 1.00

Flag question

Which genetically inherited disorder is associated with a lack of sphingomyelinase?

Select one:

- Hurler's disease
- Wilson's disease
- Kluver Bucy syndrome
- Huntington's disease

- Niemann Pick disease

Your answer is incorrect.

Niemann-pick disease is a genetically inherited disease caused by a deficiency in the enzyme sphingomyelinase. It leads to accumulation of sphingomyelin in liver, spleen, lungs, bone marrow and brain leading to irreversible neurological damage. Type A occurs in infants and type B occurs just before teenage.

The correct answer is: Niemann Pick disease

Question **76**

Not answered

Marked out of 1.00

Flag question

Which syndrome is associated with a deletion on chromosome 15?

Select one:

- Downs syndrome
- Prader Willi syndrome
- Di George's syndrome
- Cri du chat syndrome
- Edwards syndrome

Your answer is incorrect.

Two of the conditions (Angelman syndrome and Prader-Willi syndrome) involve a loss of gene activity in the same part of chromosome 15, the 15q11.2-q13.1 regions.

The correct answer is: Prader Willi syndrome

Question **77**

Not answered

Marked out of 1.00

Flag question

Which of the following structures are employed in processing emotionally salient memories?

Select one:

- Cerebellum
- Prefrontal cortex
- Amygdala
- Basal ganglia
- Angular gyrus

Your answer is incorrect.

The amygdala and hippocampal complex, two medial temporal lobe structures, are linked to two independent memory systems, each with unique characteristic functions. In emotional situations, these two systems interact in subtle but important ways. Specifically, the amygdala can modulate both the encoding and the storage of hippocampal-dependent memories. The hippocampal complex, by forming episodic representations of the emotional significance and interpretation of events, can influence the amygdala response when emotional stimuli are encountered. Ref: Phelps EA. Human emotion and memory: interactions of the amygdala and hippocampal complex. Current opinion in neurobiology.

2004 Apr 30;14(2):198-202.

The correct answer is: Amygdala

Question **78**

Not answered

Marked out of 1.00

Flag question

A gentleman presents with a 1-day history of sudden onset symptoms of akinetic mutism. What is the vascular origin of this presentation?

Select one:

- Middle cerebral artery
- Posterior cerebral artery
- Anterior cerebral artery
- Basilar artery
- Posterior inferior cerebellar artery

Your answer is incorrect.

Bilateral infarct involving the anterior cerebral artery produces quadriparesis (legs weaker than arms) and akinetic mutism (ventromedial or cingulate syndrome). Ref: Freemon FR. Akinetic mutism and bilateral anterior cerebral artery occlusion. Journal of Neurology, Neurosurgery & Psychiatry. 1971 Dec 1;34(6):693-8.

The correct answer is: Anterior cerebral artery

Question **79**

Not answered

Marked out of 1.00

Flag question

Which of the following brain regions is associated with thermoregulation?

Select one:

- Pontine nucleus
- Pre-optic area
- Ventral striatum
- Prefrontal cortex
- Medial temporal lobe

Your answer is incorrect.

The pre-optic area is a region of the hypothalamus. It is responsible for thermoregulation. It receives nervous stimulation from receptors in the skin, mucous membranes and hypothalamus Ref: Boulant JA. Role of the preoptic-anterior hypothalamus in thermoregulation and fever. Clinical infectious diseases. 2000 Oct 1;31(Supplement 5):S157-61.

The correct answer is: Pre-optic area

Question **80**

Not answered

Marked out of

John is a 67-year old man admitted to the geriatric ward with a history of recent weight loss. On obtaining further history, he admits to loss of the sense of smell. This feature is suggestive of

Select one:

1.00

Flag question

- Temporal lobe pathology
- Frontal lobe pathology
- Cerebellar pathology
- Parietal lobe pathology
- Occipital lobe pathology

Your answer is incorrect.

Anosmia is the inability to perceive odour or loss of the sense of smell. A permanent loss of smell may be caused either by a loss of olfactory receptor neurons in the nose or a damage to the olfactory nerve or the brain areas that process smell, usually after head injury or due to tumors of the frontal lobe.

The correct answer is: Frontal lobe pathology

Question **81**

Not answered

Marked out of 1.00

Flag question

The 'Pulvinar sign' seen in patients with CJD is seen in which part of the brain?

Select one:

- Pons
- Thalamus
- Cerebellum
- Striatum
- Basal ganglia

Your answer is incorrect.

Variant Creutzfeldt-Jakob disease (vCJD) is a rare cause of dementia and is causally linked to bovine spongiform encephalopathy. Symmetrical hyperintensity in the pulvinar (posterior) nuclei of the thalamus (pulvinar sign) on brain MR images was described as a specific, noninvasive, diagnostic sign of vCJD. Ref: Young GS, Geschwind MD, et al. American Journal of Neuroradiology. 2005 Jun 1;26(6):1551-62.

The correct answer is: Thalamus

Question **82**

Not answered

Marked out of 1.00

Flag question

The hormone melatonin is synthesized in which part of the brain?

Select one:

- Caudate nucleus
- Pineal gland
- Anterior pituitary
- Posterior pituitary
- Hypothalamus

Your answer is incorrect.

The pineal gland synthesizes and secretes melatonin, which has the ability to entrain biological rhythms and has important effects on reproductive function of many animals. The hormone can be used as a sleep aid and in the treatment of some sleep disorders.

The correct answer is: Pineal gland

Question **83**

Not answered

Marked out of 1.00

Flag question

The comprehension centre for language in the human brain is located on

Select one:

- Inferior temporal region
- Posterior superior temporal region
- Inferior frontal region
- Medial parietal region
- Dorsal lateral prefrontal region

Your answer is incorrect.

The superior temporal gyrus includes Wernicke's area, which is the major area involved in the comprehension of language.

The correct answer is: Posterior superior temporal region

Question **84**

Not answered

Marked out of 1.00

Flag question

Regarding CSF which of the following is true,

Select one:

- CSF has no glucose
- Glucose concentration in the CSF is same as the blood glucose concentration
- Glucose concentration in the CSF is lower than the blood glucose concentration
- Presence of glucose in CSF indicates inflammation
- Glucose concentration in the CSF is higher than the blood glucose concentration

Your answer is incorrect.

In the presence of bacterial meningitis the CSF glucose levels drop due to glycolysis by both white cells and the pathogen and impaired CSF glucose transport. The normal ratio of CSF to serum glucose as 0.6, i.e., CSF has 60% of the gucose concentration seen in the blood plasma. Ref: Nigrovic LE, et al. New England Journal of Medicine. 2012 Feb 9;366(6):576-8.

The correct answer is: Glucose concentration in the CSF is lower than the blood glucose concentration

Question **85**

Not answered

Marked out of

Corpus callosum is supplied mostly by

Select one:

1.00

Flag question

- Anterior cerebral artery
- Ascending spinal artery
- Posterior cerebral artery
- Middle cerebral artery
- Vertebral artery

Your answer is incorrect.

Infarcts of the corpus callosum are very rare; this is due to a rich blood supply from three main arterial systems: (1) the pericallosal branch of the anterior cerebral artery is most often the main vascular supply to the body (2) the subcallosal and medial callosal arteries, branches of the anterior communicating artery, provide the main supply for the anterior portion of the corpus callosum (3) The posterior pericallosal artery, a branch of the posterior cerebral artery, supplies the splenium. Ref: http://www.ncbi.nlm.nih.gov/pmc/articles/PMC4152080/

The correct answer is: Anterior cerebral artery

Question **86**

Not answered

Marked out of 1.00

Flag question

Sally is being treated for a major depressive episode with Phenelzine for the last few months. Her GP calls you to inform you that she has been diagnosed with a medical disorder and asks if she should continue the antidepressant. You advise that she should stop the antidepressant. Which illness from the list below is most likely to have prompted this advice?

Select one:

- Addison's disease
- Phaeochromocytoma
- Breast cancer
- Diabetes
- Cushing's disease

Your answer is incorrect.

The use of nonspecific monoamine oxidase inhibitors (MAOIs) like Phenelzine is contraindicated in patients with phaeochromocytoma or other tumors of the adrenal medulla, such as some neuroblastomas, that secrete pressor substances. Nonspecific MAOIs inhibit the breakdown of pressor amines, the accumulation of which can precipitate hypertensive crises. Intracranial hemorrhage and death have resulted in some cases. Ref: Eisenhofer G, Rivers G, Rosas AL, Quezado Z, Manger WM, Pacak K. Adverse drug reactions in patients with phaeochromocytoma. Drug safety. 2007 Nov 1;30(11):1031-62.

The correct answer is: Phaeochromocytoma

Question **87**

Not answered

Marked out of 1.00

Flag

Which of the following is a common side effect of a phosphodiesterase (type 5) inhibitor?

Select one:

- Priapism
- Headache

question

- Postural Hypotension
- Galactorrhoea
- Constipation

Your answer is incorrect.

A phosphodiesterase type 5 inhibitor blocks the degradation of cyclic GMP in the smooth muscle cells lining the blood vessels supplying the corpus cavernosum of the penis. The prototype drug in this class is sildenafil; other available drugs include tadalafil and vardenafil. These drugs are used in the treatment of erectile dysfunction. Headache is a common side effect of this class of medication, seen in 10% of patients. Other class-specific side effects include headache, flushing, nasal congestion, dyspepsia and muscle aches due to vasodilatory effects on the capillaries. Ref: Huang SA, Lie JD. Phosphodiesterase-5 (PDE5) inhibitors in the management of erectile dysfunction. Pharmacy and Therapeutics. 2013 Jul;38(7):407

The correct answer is: Headache

Question **88**

Not answered

Marked out of 1.00

Flag question

Sigma receptors are associated with

Select one:

- Pain
- Personality
- Dementia
- Anti-craving
- Depression

Your answer is incorrect.

Sigma (σ) receptors, initially described as a subtype of opioid receptors, are now considered unique receptors. Pharmacological studies have distinguished two types of σ receptors, termed σ1 and σ2. Of these two subtypes, the σ1 receptor has been cloned in humans and rodents, and its amino acid sequence shows no homology with other mammalian proteins. Several psychoactive drugs show high to moderate affinity for σ1 receptors, including the antipsychotic haloperidol, the antidepressant drugs fluvoxamine and sertraline, and the psychostimulants cocaine and methamphetamine; in addition, the anticonvulsant drug phenytoin allosterically modulates σ1 receptors. Recently selective σ–receptor agonists were shown to produce antidepressant-like effects in mice. Ref: Cobos EJ, Entrena JM, Nieto FR, Cendan CM, Pozo ED. Pharmacology and therapeutic potential of sigma1 receptor ligands. Current neuropharmacology. 2008 Dec 1;6(4):344-66.

The correct answer is: Depression

Question **89**

Not answered

Marked out of 1.00

Flag question

In Alzheimer's which of the following is true

Select one:

- Increased acetylcholine with increased glutamate levels
- Increased acetylcholine with increased serotonin

- Decreased acetylcholine with increased glutamate levels
- Increased acetylcholine with increased glutaminase
- Decreased acetylcholine with decreased serotonin

Your answer is incorrect.

Explanation:

The cholinergic hypothesis of Alzheimer disease (AD) is predicated on the observation that the cognitive deterioration associated with the disease results from progressive loss of cholinergic neurons and decreasing levels of acetylcholine in the brain. Cholinergic dysfunction is not the only problem in AD, and there is progressive neurodegeneration of both cholinergic and glutamatergic circuits. Glutamate has been hypothesised to be released in excess once AD develops, perhaps in part triggered by neurotoxic amyloid-beta plaques and neurofibrillary tangles that release glutamate from normal inhibition by GABA as GABA interneurons degenerate.

Ref: Taylor DM, Barnes TRE, Young AH. The Maudsley Prescribing Guidelines in Psychiatry, 14th Edition. 2021. p. 604.

Stahl SM. Stahl's Essential Psychopharmacology: Neuroscientific Basis and Practical Applications, 5th Edition. 2021. p. 515.

The correct answer is: Decreased acetylcholine with increased glutamate levels

Question **90**

Not answered

Marked out of 1.00

Flag question

Detection of 14-3-3 proteins in the CSF supports the diagnosis of

Select one:

- Pick's disease
- Binswanger's disease
- Alzheimer's disease
- Parkinson's disease
- Creutzfeldt-Jakob disease

Your answer is incorrect.

The analysis of 14-3-3 proteins in the cerebrospinal fluid (CSF) was shown to be highly sensitive and specific for the diagnosis of Creutzfeldt-Jakob disease (CJD). In sporadic CJD, this test is helpful to support the diagnosis.

The correct answer is: Creutzfeldt-Jakob disease

Question **91**

Not answered

Marked out of 1.00

Flag question

HIV is spread to brain primarily through

Select one:

- Direct entry via blood-brain barrier
- CD4 infected T-lymphocytes
- Vascular endothelial cells
- Upstream migration via axon terminals

- Infected macrophages

Your answer is incorrect.

The major HIV-1 receptors are CD4 and CD8; various chemokine receptors e.g. CXCR4 and CCR5 are considered as HIV-1 co-receptors. CD4+ helper T lymphocytes are the major routes of multiplication and entry, apart from monocytes in the peripheral blood. In the CNS, the isolated strains show preferential macrophage-tropism; further, the perivascular macrophages and microglia are the most commonly HIV- affected cells in the human brain.

The correct answer is: Infected macrophages

Question **92**

Not answered

Marked out of 1.00

Flag question

Mr. P, a patient attending your clinic, was started on clozapine a few weeks ago. He now complains of shortness of breath, tachycardia and fever. Which of the following would you request urgently?

Select one:

- Troponin
- D-dimer
- CPK
- Chest x-ray
- ECG

Your answer is incorrect.

The most important consideration is clozapine-induced myocarditis. It occurs within 6-8 weeks of starting clozapine (median 3 weeks). Symptoms include fever, tachycardia, flu-like symptoms, fatigue, dyspnea (with increased respiratory rate) and chest pain. One proposed monitoring protocol recommends obtaining baseline troponin I/T, C-reactive protein and echocardiography, and monitoring troponin and C-reactive protein on days 7, 14, 21 and 28. At least twice the upper limit of normal troponin was found in 90% of cases. In addition, persistent abnormally high heart rate or signs or symptoms consistent with infective illness should be followed by daily troponin and C-reactive protein investigation until features resolve. Ref: Ronaldson et al., Aust N Z J Psychiatry. 2011 Jun;45(6):458-65. http://www.ncbi.nlm.nih.gov/pubmed/21524186

The correct answer is: Troponin

Question **93**

Not answered

Marked out of 1.00

Flag question

Prolonged QTc interval is associated with:

Select one:

- Naltrexone
- Diazepam
- Naloxone
- Methadone
- Buprenorphine

Your answer is incorrect.

Methadone alone or combined with other arrythmogenic agents may increase the likelihood of QT interval prolongation, which is associated with torsades de pointes and can be fatal. ECG monitoring is therefore recommended whilst taking methadone.

The correct answer is: Methadone

Question **94**

Not answered

Marked out of 1.00

Flag question

Sally has a diagnosis of schizophrenia and is currently taking risperidone tablets. Her prolactin level tested recently shows an elevation that is twice the value recorded prior to commencing the treatment. Which of the following drugs could be considered as the best possible add-on treatment?

Select one:

- Amisulpride
- Aripiprazole
- Olanzapine
- Sulpride
- Quetiapine

Your answer is incorrect.

For most patients with symptomatic hyperprolactinaemia, a switch to non-prolactin elevating drug is the first choice. An alternative with modest but evolving evidence is to add aripiprazole to existing treatment. Hyperprolactinaemia and related symptoms are reported to improve fairly promptly following the addition of aripiprazole. Ref: Mir et al., Psychopharmacol. 2008 May;22(3):244-53.

The correct answer is: Aripiprazole

Question **95**

Not answered

Marked out of 1.00

Flag question

A 47-year-old man developed seizures on taking an antipsychotic medication. The psychiatrist treating him initially reduced the dose of his antipsychotic, and later started him on sodium valproate. Which of the following antipsychotic drugs is most likely to produce this effect?

Select one:

- Fluphenazine
- Haloperidol
- Clozapine
- Risperidone
- Pimozide

Your answer is incorrect.

Seizures may occur at any time whilst taking clozapine. It is a dose and plasma level-related effect. Prophylactic valproate or lamotrigine is used in high-risk subjects when titrating to a high dose of clozapine. EEG abnormalities are common in those on clozapine treatment; these abnormalities do not always indicate seizure activity.

The correct answer is: Clozapine

Question **96**

Not answered

Marked out of 1.00

Flag question

Mr. P developed cardiomyopathy on taking this medication for the last 9 months. The most likely drug is

Select one:

- Clozapine
- Amisulpride
- Risperidone
- Olanzapine
- Quetiapine

Your answer is incorrect.

Clozapine is associated with myocarditis and cardiomyopathy. Myocarditis seems to occur within 6-8 weeks of starting clozapine (median 3 weeks) and cardiomyopathy may occur later in treatment (median 9 months) but both may occur at anytime.

The correct answer is: Clozapine

Question **97**

Not answered

Marked out of 1.00

Flag question

Which one among the following is a risk factor for SSRI induced hyponatraemia?

Select one:

- High initial sodium
- Cold weather
- Male sex
- Young age
- Low body mass

Your answer is incorrect.

Risk factors for SSRI induced hyponatraemia include old age, female sex, low body weight, low baseline sodium concentration, history of hyponatraemia, warm weather, reduced renal function, medical co-morbidity and some drug treatments like carbamazepine, antipsychotics, thiazide diuretics, NSAIDs, cancer chemotherapy and drugs like tramadol, omeprazole and trimethoprim.

The correct answer is: Low body mass

Question **98**

Not answered

Marked out of 1.00

Flag question

Sally has depression but would accept her treatment only if the probability of drug induced weight gain is low. Which of the following antidepressants is best to avoid?

Select one:

- Venlafaxine
- Fluoxetine
- Mirtazapine

- Sertraline
- Duloxetine

Your answer is incorrect.

Mirtazapine is associated with weight gain in around 1/5th of subjects, especially in the first 4 weeks of treatment. This effect may be secondary to its effects on the 5-HT2C and H1 receptors. Ref: Laimer M, Kramer-Reinstadler K, Rauchenzauner M, et al. Effect of mirtazapine treatment on body composition and metabolism. J Clin Psychiatry. 2006;67:421-424

The correct answer is: Mirtazapine

Question **99**

Not answered

Marked out of 1.00

Flag question

Which of the following is primarily tested by using Abnormal Involuntary Movement Scale (AIMS)?

Select one:

- Torticollis
- Akathisia
- Psychomotor agitation
- Tardive dyskinesia
- Jaw stiffness

Your answer is incorrect.

The AIMS records the occurrence of tardive dyskinesia (TD) in patients receiving neuroleptic medications. This test is used to detect TD and to follow the severity of a patient's TD over time.

The correct answer is: Tardive dyskinesia

Question **100**

Not answered

Marked out of 1.00

Flag question

Which one of the following is the least likely feature associated with Neuroleptic Malignant Syndrome?

Select one:

- Hyperthermia
- Muscle rigidity
- Autonomic lability
- Persistent bradycardia
- Raised CPK levels

Your answer is incorrect.

Bradycardia is not a feature seen in patients with NMS. The symptoms and signs of NMS include hyperthermia, rigidity, agitation, altered levels of consciousness, tachycardia, tachypnoea, hypertension, tremor, diaphoresis, raised CPK levels, leukocytosis and metabolic acidosis.

The correct answer is: Persistent bradycardia

Question **101**

Not answered

Marked out of 1.00

Flag question

The upper normal value of QTc in females is

Select one:

- 400 msec
- 470 msec
- 500 msec
- 650 msec
- 440 msec

Your answer is incorrect.

The upper normal value of QTc in males is 440 msec and females is 470 msec. Mounting evidence links QTc values over 500 msec to a clearly increased risk of arrhythmia.

The correct answer is: 470 msec

Question **102**

Not answered

Marked out of 1.00

Flag question

Which of the following drugs increases the incidence of cleft palate when used in pregnant women?

Select one:

- Gabapentin
- Lithium
- Topiramate
- Lamotrigine
- Sertraline

Your answer is incorrect.

Lamotrigine increases the incidence of cleft palate. There are some anecdotal reports linking valproate and diazepam as well to this abnormality.

The correct answer is: Lamotrigine

Question **103**

Not answered

Marked out of 1.00

Flag question

Which class of antidepressants, when used during pregnancy increases the risk of cleft lip/cleft palate in the newborn?

Select one:

- Tricyclic antidepressant
- Monoamine oxidase inhibitors
- Noradrenergic and specific serotonergic antidepressant
- Selective serotonin reuptake inhibitors
- Norepinephrine reuptake inhibitor

Your answer is incorrect.

Existing research evidence has shown a connection between prenatal use of SSRI anti-depressants and the subsequent formation of birth defects like cleft lip and palate. In 2006, a Danish population-based cohort study published in Epidemiology reported that women taking SSRI antidepressants from 30 days before conception to the end of the first trimester were more likely to give birth to babies with congenital formations, which may include cleft lip and palate.

The correct answer is: Selective serotonin reuptake inhibitors

Question **104**

Not answered

Marked out of 1.00

Flag question

Which of the following adverse effects has not been reported in association with therapeutic doses of SSRIs?

Select one:

- Galactorrhea
- Akathisia
- Hypertension
- Photosensitivity
- Hallucinations

Your answer is incorrect.

Several published reports of SSRI-induced movement disorders are available; they mostly report akathisia, dyskinesia, parkinsonism and occasionally dystonia and bruxism. Most of these were reported when using normal therapeutic doses. SSRIs, especially escitalopram and paroxetine, are occasionally associated with galactorrhea. Though rare, photosensitivity and hallucinations have been anecdotally reported with SSRIs. Nevertheless, therapeutic range of SSRI dose is not associated with systemic hypertension, though SNRIs such as venlafaxine carries this risk especially above the dose of 225mg/day. Ref: Ann Pharmacother. 1998 Jun;32(6):692-8.; Clin Exp Dermatol. 2009 Dec;34(8):e763-5.; J Clin Psychopharmacol. 2000 Oct;20(5):579-80.; Br J Clin Pharmacol. 1997;44277–81.; Journal of Clinical Psychiatry. 1996 Oct 1;57(10):449-54.

The correct answer is: Hypertension

Question **105**

Not answered

Marked out of 1.00

Flag question

Which of these drugs is likely to cause tricuspid valve displacement towards the apex of the right ventricle in the newborn, when used during pregnancy?

Select one:

- Lithium
- Clozapine
- Valproate
- Haloperidol
- Carbamazepine

Your answer is incorrect.

Ebstein's anomaly is a congenital heart defect in which the tricuspid valves are displaced towards the apex of the right ventricle. Lithium use has a well-known association with this cardiac malformation (relative risk is 10-20 times more than control). Some case reports have also implicated benzodiazepines, although this is rare.

The correct answer is: Lithium

Question **106**

Not answered

Marked out of 1.00

Flag question

The risk of developing seizures with the use of clozapine is increased by:

Select one:

- 0.6
- 0.5
- 0.1
- 0.4
- 0.2

Your answer is incorrect.

Patients can have generalized tonic-clonic seizures during treatment with clozapine. Life-table analysis predicts a cumulative 10% risk of seizures after 3.8 years of treatment. Clozapine-related seizures appear to be dose-related. High-dose therapy (greater than or equal to 600 mg/day) was associated with a greater risk of seizures (4.4%) than medium (300 to 600 mg/day; 2.7%) or low doses (less than 300 mg/day; 1.0%). Also, rapid upward titration may increase seizure risk. Ref: Devinsky O, Honigfeld G, Patin J. Clozapine-related seizures. Neurology. 1991 Mar 1;41(3):369)

The correct answer is: 0.1

Question **107**

Not answered

Marked out of 1.00

Flag question

Mr. X was started on Sertraline recently. He has now developed nausea and diarrhea. Which receptors, when perturbed by SSRIs, could cause this effect?

Select one:

- 5HT-4
- 5HT-3
- 5HT-2C
- 5HT-1
- 5HT-7

Your answer is incorrect.

The most common GI side effects of SSRIs are nausea, vomiting, diarrhea, anorexia and dyspepsia. It mostly results from stimulation of 5-HT3 receptors.

The correct answer is: 5HT-3

Question **108**

Not answered

Marked out of 1.00

Flag question

An antidepressant with good efficacy in patients with atypical depression is

Select one:

- Phenelzine
- Amitriptyline
- Fluoxetine
- Mirtazapine
- Venlafaxine

Your answer is incorrect.

Differential response to antidepressants has been used as a diagnostic feature of atypical depression; a preferential response to MAOIs compared with TCAs but a modest response to ECT characterizes this distinct subtype. Ref: Henkel V, Mergl R, Allgaier A, et al. Treatment of depression with atypical features: a meta-analytic approach. Psychiatry Res. 2006;141:89-101.

The correct answer is: Phenelzine

Question **109**

Not answered

Marked out of 1.00

Flag question

Sigma-1 receptors are associated with which of the following?

Select one:

- Anti-craving effects of naltrexone
- Pain perception
- Impulsivity
- Flashbacks
- Dissociation

Your answer is incorrect.

The sigma-1 receptor (σ1R) were initially considered as one type of opioid receptors; but later various unique properties were discovered; at present certain neurosteriods are considered to be the endogenous ligands for σ1R. They appear to modulate (1) nociceptive messages. σ1R antagonists could have a place in the treatment of pain states especially neuropathic and other chronic pain conditions. (2) PCP-induced cognitive impairment seen in animal models of schizophrenia is reversed on stimulation of sigma-1 receptor Drugs such as fluvoxamine, sertraline, haloperidol, stimulants cocaine and methamphetamine and donepezil have affinity towards σ1R; the anticonvulsant drug phenytoin allosterically modulates σ1 receptors. Via the molecular chaperone activity, the sigma-1 receptor regulates protein folding/degradation, intracellular calcium, ER/oxidative stress, and cell survival. The σ1 receptor has an uniquely human amino acid sequence that has no homology with other mammalian proteins. Certain neurosteroids are known to interact with σ1 receptors, and have been proposed to be their endogenous ligands. They also modulate the glutamatergic system. Ref: Cobos EJ, et al. Current neuropharmacology. 2008 Dec 1;6(4):344-66.

The correct answer is: Pain perception

Question **110**

Flumazenil reverses action of benzodiazepines through which of the following action

Not answered

Marked out of 1.00

Flag question

Select one:

- Noncompetitive inhibition at the GABA-B receptor
- Competitive inhibition at the GABA-A receptor
- Closing chloride channels
- Noncompetitive inhibition at the GABA-A receptor
- Closing potassium channels

Your answer is incorrect.

Flumazenil antagonizes the actions of benzodiazepines on the central nervous system. It reverses the effects of benzodiazepines by competitive inhibition at the benzodiazepine-binding site on the GABAA receptor.

The correct answer is: Competitive inhibition at the GABA-A receptor

Question **111**

Not answered

Marked out of 1.00

Flag question

Lucy had nausea and vomiting after taking fluoxetine. This effect is most likely to be due to,

Select one:

- 5HT3 stimulation
- 5HT2 stimulation
- 5HT1 stimulation
- 5HT3 antagonism
- 5HT1 antagonism

Your answer is incorrect.

SSRIs like fluoxetine can cause gastric side effects like abdominal discomfort, nausea and vomiting. This effect is due to 5HT3 agonism or stimulation

The correct answer is: 5HT3 stimulation

Question **112**

Not answered

Marked out of 1.00

Flag question

Which psychotropic drug was initially used as a preoperative antihistamine prior to general anesthesia?

Select one:

- Phenytoin
- Lithium
- Clomipramine
- Chlorpromazine
- Carbamazepine

Your answer is incorrect.

Between 1950 - 1952 presurgical antihistamine chlorpromazine was shown to have antipsychotic effects independently by Delay and Deniker's team, and Charpentier from Rhône-Poulenc in France. In 1955 Delay coined the term neuroleptic.

The correct answer is: Chlorpromazine

Question **113**

Not answered

Marked out of 1.00

Flag question

Tom is a research student studying psychopharmacology. He wants to which of the following is the most potent inhibitor of CYP2D6. The correct answer is

Select one:

- Escitalopram
- St. John's Wort
- Reboxetine
- Sertraline
- Paroxetine

Your answer is incorrect.

Paroxetine, Fluoxetine and Duloxetine are potent inhibitors of CYP2D6 enzyme system. Bupropion and Reboxetine are weak inhibitors. Ref: Sanchez et al. Int Clin Psychopharmacol. 2014 Jul;29(4):185-96

The correct answer is: Paroxetine

Question **114**

Not answered

Marked out of 1.00

Flag question

Mr. Z has a history of alcohol dependence and develops recurrent infections. His GP advised him to avoid certain antibiotics as they can cause facial flushing with alcohol. The most likely offending drug in this regard is

Select one:

- Amoxicillin
- Metronidazole
- Co-amoxiclav
- Cefuroxime
- Ciprofloxacin

Your answer is incorrect.

There is evidence that taking alcohol and metronidazole together may also cause disulfiram-like reactions. The common effects include facial flushing and a sudden drop in blood pressure. Ref: Williams CS & Woodcock KR. Ann Pharmacother. 2000 Feb;34(2):255-7 There is evidence that taking alcohol and metronidazole together may also cause disulfiram-like reactions. The common effects include facial flushing and a sudden drop in blood pressure. Ref: Williams CS & Woodcock KR. Ann Pharmacother. 2000 Feb;34(2):255-7

The correct answer is: Metronidazole

Question **115**

Not answered

Marked out of 1.00

Flag question

Mrs. P is taking Paroxetine for her depressive symptoms. She has medical co-morbidities. Which of the following must be used with caution with Paroxetine?

Select one:

- Amoxicillin
- Haloperidol
- Naproxen
- Lithium
- Quetiapine

Your answer is incorrect.

Paroxetine is a type of SSRI drug. NSAIDs and SSRIs both cause alterations of thrombocyte aggregation and consequently increase the risk of gastrointestinal adverse effects . Examples of NSAIDs include Ibuprofen, Diclofenac, Naproxen, Indomethacin, Aceclofenac and Piroxicam.

The correct answer is: Naproxen

Question **116**

Not answered

Marked out of 1.00

Flag question

Paul is on psychotropic medication and decides to quit smoking. When he did, he experienced severe sedation. Which of the following medication is he most probably taking?

Select one:

- Aripiprazole
- Clozapine
- Risperidone
- Quetiapine
- Olanzapine

Your answer is incorrect.

Patients who smoke but are stabilized on clozapine will require a dose adjustment (reduction) when they quit smoking, as the CYP1A2 inducing effects of polycyclic aromatic hydrocarbons is removed on abstaining from smoking. If the dose is not reduced, patients may experience dose-dependent adverse effects , especially tachycardia, sedation, weight gain and seizures. Ref: Smith TL, Mican LM. What to do when your patient who takes clozapine enters a smoke-free facility. Current Psychiatry. 2014 May 1;13(5):55.

The correct answer is: Clozapine

Question **117**

Not answered

Marked out of 1.00

Flag

Which of the following drugs can cause an increase in serum lithium levels and should not be co-prescribed with it?

Select one:

- Olanzapine
-

question

Captopril

- Venlafaxine
- Mirtazapine
- Sodium valproate

Your answer is incorrect.

Captopril is a type of ACE inhibitor (angiotensin converting enzyme inhibitor). It can reduce thirst, which can lead to mild dehydration and increase renal sodium loss, leading to increased sodium reabsorption by the kidney, causing an increase in plasma lithium levels. ACE inhibitors can cause up to 4-fold increase in plasma concentration of lithium and could precipitate lithium toxicity.

The correct answer is: Captopril

Finish review

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Fresh Practice Questions 2017 Set 2

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